Software Requirements Specification

Open Clinical Record

Open Clinical Record Internship Project

MVP Scope Revision — September 2026

Document Status: Revised MVP Baseline

Document Item	Value
Product	Open Clinical Record
Document	Software Requirements Specification (SRS)
Version	1.4 — internship MVP scope revision
Status	Revised MVP implementation baseline
MVP scope	Patient Management, Patient Chart, and Appointment Management
Primary users	Clinician/Doctor; Nurse/Clinical Staff; Receptionist/Front Desk
Implementation period	Remaining one-month internship period
Standards	International interoperability standards are deferred from the MVP
Future functionality	Additional EMR capabilities may be considered after the core MVP is complete

This document intentionally limits the internship MVP to functionality that can realistically be implemented, tested, and demonstrated within the remaining project period.

Contents

1 Introduction

1.1 Purpose

This Software Requirements Specification defines the reduced and prioritized MVP for Open Clinical Record (OCR). The scope is limited to functionality that can be implemented and tested within the remaining one-month internship period.

The MVP is focused on three core business areas: **Patient Management**, **Patient Chart**, and **Appointment Management**. The objective is to deliver these workflows as a coherent, working system before considering additional functionality.

1.2 Scope of Product

Open Clinical Record is a focused outpatient patient-management and appointment system with a basic electronic patient chart. The patient is the central record around which the three MVP areas are connected.

Patient Management → Patient Chart → Appointment Management → Check-in/Visit History

The MVP does not attempt to implement a complete hospital EMR. Full clinical encounter documentation, diagnosis, treatment, prescriptions, advanced medical reports, laboratory, pharmacy, billing, radiology, patient portal, advanced analytics, and standards-based interoperability are deferred.

1.3 Definitions

Term	Definition
EMR	Electronic Medical Record; a digital record containing patient and healthcare-related information.
MVP	Minimum Viable Product; the smallest planned version that satisfies the approved internship scope.
Patient Management	Functions for registering, identifying, searching, viewing, and maintaining basic patient information.
Patient Chart	A longitudinal view of information associated with a patient, including demographics, allergies, medication/history information, alerts where required, and appointment/visit history.
Appointment	A planned date/time for a patient's scheduled visit.
Appointment status	The current state of an appointment, such as scheduled, checked-in, cancelled, completed, or other approved states.
Visit	An occurrence in which a patient attends the clinic; it may originate from an appointment or a basic walk-in.
Walk-in	A patient visit that begins without a previously scheduled appointment.
RBAC	Role-Based Access Control; access based on the authenticated user's assigned application role.

Audit trail	A record of important system actions including actor and time information.

1.4 References

The requirements are informed by the project's existing research and discovery artifacts, including clinical workflow, patient-chart research, appointment-scheduling research, and the approved project scope.

- `docs/00-project/project-scope.md`
- `docs/02-discovery/clinical-workflow.md`
- `docs/01-research/patient-chart-research.md`
- `docs/01-research/appointment-scheduling-research.md`
- `docs/03-requirements/non-functional-requirements.md`
- `docs/04-architecture/architecture-overview.tex`

2 General Description

2.1 Product Perspective

The system is designed around a simple clinic workflow. Staff first identify or register the patient, then access the patient chart, manage an appointment, and record attendance/check-in information. The resulting appointment and visit history remains associated with the patient.

The system should be modular enough to support future expansion, but future capabilities must not delay completion of the three approved MVP areas.

2.2 MVP Modules

1. **Patient Management**: patient registration, unique identifier, search, profile, demographic/contact information, and basic status.
2. **Patient Chart**: patient context, allergies, relevant medication/history information, alerts where required, and appointment/visit history.
3. **Appointment Management**: appointment creation, viewing, rescheduling, cancellation, status, check-in, and basic walk-in support.

Authentication, authorization, validation, error handling, and basic audit logging are cross-cutting support concerns. They are not additional business modules.

2.3 User Characteristics

The MVP has exactly three application roles.

Role	Responsibility	MVP capabilities
Receptionist / Front Desk	Patient registration and scheduling	Register/search patients; maintain permitted demographics; create, view, reschedule and cancel appointments; check in patients; support basic walk-ins.
Nurse / Clinical Staff	Clinical support	Search/view patients and charts; view appointments; support check-in/visit workflow; maintain permitted chart information according to the approved permission matrix.
Clinician / Doctor	Clinical oversight and patient care context	Search/view patient charts; review patient history and appointments; perform authorized clinical/chart updates required by the MVP.

No System Administrator or other application role is included in the MVP.

2.4 Constraints

- The MVP must remain small enough to implement and test within the remaining one-month internship period.
- Only three business modules are committed: Patient Management, Patient Chart, and Appointment Management.
- Exactly three application roles are used: Receptionist/Front Desk, Nurse/Clinical Staff, and Clinician/Doctor.
- International healthcare standards and FHIR integration are not required for the internship MVP.
- Additional functionality may be added only after the core MVP is complete and only if time remains.
- Patient information must be protected from unauthorized access.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interface

The system shall provide clear screens for authentication, patient search, patient registration, patient profile/chart, appointment management, appointment details, check-in, and basic visit history.

The interface shall show the current patient context clearly and shall not present controls for operations the current user is not authorized to perform.

3.1.2 Software Interface

The React frontend shall communicate with the ASP.NET Core backend through the REST API defined by the architecture. The backend shall enforce validation and authorization independently of the frontend.

3.1.3 Communication

Client/server communication shall use secure transport in the deployed application. Communication or save failures shall be reported clearly and shall not falsely indicate successful persistence.

3.2 Functional Requirements

3.2.1 Patient Management Requirements

ID	Requirement	Detailed specification
FR-PM-001	**Register patient**	The system shall allow an authorized Receptionist/Front Desk user to create a patient record with the required demographic and contact information.
FR-PM-002	**Unique patient ID**	The system shall assign or record a unique patient identifier for each patient.
FR-PM-003	**Search patient**	The system shall allow authorized users to search for a patient using supported identifying information.
FR-PM-004	**View patient profile**	The system shall display the patient's basic demographic, contact, and status information to authorized users.
FR-PM-005	**Update patient information**	The system shall allow authorized users to update permitted patient information while preserving the integrity of the patient record.
FR-PM-006	**Prevent duplicate patients**	The system shall validate identifying information and provide appropriate feedback when a likely duplicate patient is detected.
FR-PM-007	**Patient status**	The system shall maintain a basic patient status and apply approved restrictions to actions affected by that status.

3.2.2 Patient Chart Requirements

ID	Requirement	Detailed specification
FR-PC-001	**Open patient chart**	The system shall allow an authorized user to open a patient's chart from a valid patient record.
FR-PC-002	**Chart summary**	The chart shall provide a concise view of important patient information available within the MVP.

FR-PC-003	**Allergy information**	The system shall allow appropriately authorized users to record and view allergy information associated with the correct patient.
FR-PC-004	**Medication/history information**	The system shall allow appropriately authorized users to record and view relevant medication/history information associated with the patient.
FR-PC-005	**Patient alerts**	The system shall support recording and displaying important patient alerts where required by the approved workflow.
FR-PC-006	**Appointment history**	The patient chart shall allow authorized users to view the patient's appointment history.
FR-PC-007	**Visit/check-in history**	The system shall preserve and display basic visit or check-in history associated with the patient.
FR-PC-008	**Correct-patient association**	The system shall ensure chart information is associated with the intended patient and shall prevent accidental cross-patient updates.

3.2.3 Appointment Management Requirements

ID	Requirement	Detailed specification
FR-AP-001	**Create appointment**	The system shall allow an authorized Receptionist/Front Desk user to create an appointment for an existing patient.
FR-AP-002	**View appointments**	Authorized users shall be able to view appointments relevant to their work.
FR-AP-003	**Appointment details**	Each appointment shall identify the patient, scheduled date/time, purpose/reason where required, and current status.
FR-AP-004	**Reschedule appointment**	The system shall allow authorized users to change an eligible appointment's scheduled date/time while retaining sufficient history.
FR-AP-005	**Cancel appointment**	The system shall allow authorized users to cancel an eligible appointment and retain its historical existence.
FR-AP-006	**Appointment status**	The system shall maintain a clear appointment status throughout its lifecycle.
FR-AP-007	**Check-in**	The system shall allow Receptionist/Front Desk or Nurse/Clinical Staff to record patient check-in for an eligible appointment.
FR-AP-008	**Basic walk-in**	The system shall support a basic walk-in path without requiring a fabricated prior appointment.
FR-AP-009	**Visit linkage**	Check-in/visit information shall remain linked to the correct patient and, when applicable, the originating appointment.
FR-AP-010	**Appointment history**	The system shall preserve meaningful appointment history when an appointment is rescheduled or cancelled.

3.2.4 Authentication, Authorization, Validation and Audit

ID	Requirement	Detailed specification
FR-SEC-001	**Authenticate user**	The system shall authenticate users before granting access to protected functions or patient information.
FR-SEC-002	**Enforce three roles**	The system shall enforce permissions for exactly three MVP roles: Receptionist/Front Desk, Nurse/Clinical Staff, and Clinician/Doctor.
FR-SEC-003	**Restrict access**	The system shall deny operations for which the authenticated user lacks permission.
FR-SEC-004	**Audit important actions**	The system shall record important patient and appointment actions with actor and timestamp information where required.
FR-VAL-001	**Validate data**	The system shall validate required fields, formats, identifiers, relationships, and relevant business rules before saving data.
FR-VAL-002	**Clear errors**	The system shall provide understandable error feedback without exposing unnecessary protected information.
FR-VAL-003	**Safe failure**	A failed operation shall not silently create partial or inconsistent patient/appointment records.
FR-VAL-004	**Preserve consistency on failure**	The system shall prevent failed operations from leaving incomplete or inconsistent related records when the operation requires atomic persistence.

3.3 Business Rules

ID	Business rule
BR-001	Each patient has one unique patient identifier within the system.
BR-002	Patient chart information must belong to the correct patient.
BR-003	Cancelled appointments remain in history rather than being silently deleted.
BR-004	Rescheduling shall not erase the fact that an earlier appointment state existed.
BR-005	A walk-in may create a visit without a prior appointment.
BR-006	Appointment operations must respect the current patient and appointment status.
BR-007	Chart access does not automatically grant permission to modify every type of chart information.
BR-008	Only the three approved application roles may be assigned to MVP users.
BR-009	Important patient and appointment actions should be auditable.
BR-010	Additional EMR functionality is deferred until the three core modules are complete.

3.4 Non-Functional Requirements

The existing detailed NFR document remains supporting material, but the internship MVP prioritizes the following qualities:

- **Usability:** common patient and appointment tasks should be straightforward for clinic staff.
- **Performance:** ordinary patient search and appointment operations should respond promptly under expected internship-scale usage.
- **Security:** authenticated users must only access permitted patient information and operations.
- **Data integrity:** patient and appointment records must not be silently corrupted or lost.
- **Reliability:** failed operations should produce safe, understandable outcomes.
- **Maintainability:** the implementation should keep the three modules separated enough for future extension.

3.5 Logical Database Requirements

The database shall support the three MVP modules without introducing unnecessary enterprise complexity.

Patient Management: Patient.

Patient Chart: Allergy, Medication History, Patient Alert, and patient-linked appointment/visit history.

Appointment Management: Appointment, Appointment History/Event, Visit.

Cross-cutting: User, Audit Event.

The central relationships are Patient 1–M Appointment, Patient 1–M Visit, and Appointment 0..1–1 Visit where a visit originated from an appointment. A walk-in visit shall be allowed without an appointment reference.

The current logical ERD in `docs/03-requirements/ER/OCR_MVP_ERD.html` defines the MVP entity set and relationships. The implementation shall use the simplest normalized model that supports the approved MVP.

3.6 Deferred Functionality

The following are explicitly outside the committed MVP and shall not be used to expand the implementation schedule:

- Full encounter documentation
- Diagnosis and treatment documentation
- Prescription management
- Advanced medical report/document lifecycle
- Document amendments/versioning
- Laboratory, pharmacy, billing, insurance, and radiology modules

- Patient portal/mobile application
- SMS and external integrations
- FHIR or other international standards implementation
- Advanced analytics and enterprise scheduling
- AI clinical decision support

These may be considered as future work only after the MVP is functional and tested.

3.7 Acceptance Criteria

The MVP is considered ready for demonstration when the following can be completed successfully:

1. A receptionist can register a patient and receive a unique patient ID.
2. An authorized user can search for and open that patient.
3. The patient's chart displays the stored patient information and relevant history.
4. Authorized staff can record and view supported chart information.
5. A receptionist can create an appointment for the patient.
6. Authorized staff can view the appointment and update its status/check in the patient.
7. An appointment can be rescheduled or cancelled without silently deleting its history.
8. A basic walk-in can be recorded without inventing a previous appointment.
9. Role restrictions prevent unauthorized operations.
10. Invalid input and failed operations are handled without creating inconsistent records.

3.8 Traceability Summary

MVP area	Main requirements
Patient Management	FR-PM-001 through FR-PM-007
Patient Chart	FR-PC-001 through FR-PC-008
Appointment Management	FR-AP-001 through FR-AP-010
Cross-cutting controls	FR-SEC-001 through FR-SEC-004; FR-VAL-001 through FR-VAL-004

3.9 Open Validation Items

The following decisions should be confirmed before the requirements are considered frozen:

- Mandatory patient registration fields.
- Exact permissions for each of the three roles.

- Required allergy, medication/history, and alert fields.
- Appointment statuses required for the MVP.
- Whether basic walk-in support is required for the final demonstration or can be deferred.
- Required level of audit logging.

4 Scope Change Control

Any new feature proposed during project work should first be classified as either MVP or Future. A Future feature shall not displace unfinished core functionality unless the scope change is formally approved.